

‘There’s more to it than allure...’ – Navigating Socio-cultural Roles of Digital Jewellery

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ABSTRACT

Over the recent years, digital jewellery has been creatively explored in research and commercial products. However, the symbolic layers of these artefacts which are inherent to traditional jewellery are not yet understood. Exploring the anthropological dimensions of wearable ornaments is necessary for digital jewellery to become enduring assets of personal importance. Drawing on participatory design methods, we conducted an explorative workshop with experts in design, art, and technology (N=16) discussing the socio-technical dimensions of designing interactive jewellery. In this work, we build upon a multidisciplinary framework describing traditional jewellery from an anthropological perspective. We examine whether and how novel digital ornaments fulfil five major socio-cultural roles of jewellery. Our work contributes insights for designing digital jewellery with social relevance and hints new directions for future implementations.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Digital Jewellery, Wearable Computing, Participatory Design, Interactive Jewellery

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1 INTRODUCTION

Along with increasing ubiquity of technology, its social perception has been shifting from purely pragmatic and function-oriented to more symbolic or even metaphorical [29]. This phenomenon is especially visible in wearable technology which, as the part of human apparel, has the potential to satisfy all the roles that fashion and accessories have served from the very beginning of human history. Among different garments, jewellery offers a particularly interesting design space, as a traditional carrier of social and cultural expressions. Even though state-of-art technology can easily be incorporated to enhance common form factors, digital jewellery remains a niche.

Past research in HCI developed a rich body of work on designing wearable technologies [15, 20, 40], yet the symbolic and cultural meaning of particular garments have gathered limited attention. The incorporation of electronics in everyday garments impel the need to understand not only ergonomic and functional requirements for such creations, but also social, cultural and emotional implications of wearing them [10]. In our work, we aim to explore the dimensions of designing interactive jewellery through the lens of socio-cultural roles that jewellery traditionally served. We conducted three workshops with design experts, focused on exploring (1) how to design digital jewellery to satisfy different roles of traditional ornaments, and (2) which interactive functionalities can effectively support each of these roles and why. In this work, we discuss the results from these workshops and formulate actionable takeaways for designing digital jewellery. Our work further investigates the socio-technical aspects of interactive garments from a user-centred perspective. In particular, we aim to build a deeper understanding of users’ perception and sentiments related to

interweaving technology into ornamental accessories. These investigations will facilitate future efforts in designing socially-relevant wearable technologies.

2 RELATED WORK

First, we review past works that explored the concept of digital jewellery from various perspectives. Then, we introduce the framework for describing traditional jewellery that our analytical lens is built upon.

2.1 Interweaving Technology and Jewellery

While interactive jewellery has limited presence in the consumer market, the concept of enhancing ornaments with technology has been widely explored in research and artistic work. A notable trend of “seamless integration of technology into jewellery” has been defined by Miner et al. [30], who advocated for embedding different digital functionalities in an unobtrusive way, hiding them in jewellery-looking devices. Their viewpoint inspired a number of works that used jewellery form factors to explore various types of reminders [11], sleep [26] or mood [4] trackers, notifications [14], or implemented multi-function wearable devices mimicking the appearance of popular jewellery items [12]. These works provide insights on designing wearables that utilize form factors of ornamental garments, yet they do not aim at supporting the socio-cultural dimensions of jewellery. In our work, we establish a holistic view on designing interactive jewellery, in ways that preserve not only the aesthetics, but also symbolic, cultural, and emotional dimensions of traditional jewellery.

How digital devices can fulfil and expand the roles served by traditional jewellery gathered limited attention in HCI. Different electronic-based embodiments of jewellery were explored by Versteeg et al. [42], highlighting the role of interaction between the user and the particular piece of jewellery. Arora et al. [5] report that the interaction with digital jewellery reaches beyond the wearer, sometimes actively involving spectators and bystanders. The ability to act as a subtle communication channel has already been used in the Victorian era, e.g. in the form of acrostic rings [43]. Nowadays, this context of use is further enhanced to support intimate communication between couples [22] or close friends [1, 23]. Our inquiry aims at understanding how particular design decisions and features affect the sensitive balance between functional, aesthetic, cultural, emotional and communicative aspects of interactive jewellery.

2.2 Jewellery in Context

In our work, we look at modern interactive jewellery through the lens of socio-cultural aspects of traditional jewellery. We look at past research in cultural anthropology to inform our inquiry [2, 16]. In particular, our considerations are anchored in the framework formulated by Marjan Unger [41]. The majority of studies on traditional jewellery focus primarily on pieces crafted with precious metals [8] or expensive designer jewellery [18], putting aside relatively new forms of mass-produced fashion jewels or items worn by various social groups to identify and express themselves. Unger’s framework presents a holistic approach to study jewellery, comparing its values within the context of cultural anthropology, navigating psychological and social implications of (not) fulfilling various roles

that jewellery serves for its wearers. They render their definition of jewellery as *an object that is worn on the human body, as a decorative and symbolic addition to its outward appearance* [41, p.18]. ‘Symbolic’ refers to various meanings comprised in a piece of jewellery, including the reasons of wearing it. Unger derived five fundamental roles of wearing jewellery [41, p.16]:

- **adornment** – embracing the human joy of decoration,
- **identity** – demarcating one’s identity and revealing grains of personality to others,
- **memento** – keeping memories alive by materialising them as a wearable object,
- **prosperity** – showing one’s position within society,
- **protection** – mitigating fears and protecting oneself from any kind of (imaginary) threat.

In the remainder of this paper, we will refer to those five dimensions as roles, motivations or values, following the context of their referral. While the design of jewellery tends to express a complex blend of various motivations, many pieces reveal which motivations prevailed in the creator’s or wearer’s perception. It is vital for modern interactive jewellery to address these dimensions using new metaphors and modalities. In our inquiry, we aim to explore how different features and designs of interactive jewellery can facilitate fulfilling various motivations.

3 METHOD

To navigate the design challenges and opportunities of interactive ornaments serving the traditional socio-cultural roles of jewellery, we organised a series of creative workshops with expert audiences. We used participatory techniques that have proven efficient in exploring vague or fuzzy challenges involving technology [34, 37]. Blending perspectives during the workshops allows for triangulation [3, 35], contributing to more credible and trustworthy inquiry [17]. Past studies on wearables [25], decorative garments [12], and smart jewellery [19] proved this method efficient for designing artefacts.

3.1 Participants

Since the digital jewellery is not yet widely adopted and its exposure to general audience is limited, developing novel design strategies and concepts might be too difficult for non-experts. Therefore, we decided to conduct our workshop with expert participants instead of potential users. We recruited participants with design-related backgrounds: (1) fashion and jewellery designers, (2) HCI-master’s students, (3) industrial design majors, and (4) amateur jewellery and arts enthusiasts. Recruiting an expert and semi-expert audience enabled us to gather insights that reflect needs that users may be unaware of, and to draw from the multitude of perspectives the experts are aware of [31]. We conducted three design sessions with a total of 16 experts (4 male, 12 female, aged 18–33, $M = 22.94$, $SD = 4.04$). All attendees were served a warm lunch as gratuity for their participation.

3.2 Procedure

As our workshops did not aim at developing complete designs, we deliberately blended the expertise of participants facilitating group discussions on various dimensions of designing jewellery. This way,

we benefited from the constructivist paradigm [13, 38], advocating that new ideas are determined by one's past experiences.

Each workshop session lasted approx. 1h 30m and began with introducing the topic and schedule. We asked all participants to sign consent forms and to answer a short open-ended survey on their experience with design at large, interest in jewellery, and technological proficiency. We began each workshop with a 15-minute talk that covered the definition of jewellery and its roles along the framework by Unger [41] (cf. section 2.2), and presenting examples of digital jewellery pieces to inspire the creative process [39]. The presentation part concluded with an open discussion about how the participants understand different axes of Unger's framework.

The remainder of the session consisted of two brainstorming phases on particular design challenges: (Q1) What digital jewellery should be like to support: adornment, identity, memento, prosperity, and protection?; and (Q2) Which digital features can be used to support each of the 5 roles?. Q1 was intentionally imprecise to allow for different creative associations not limited to one aspect of design. Both phases employed rapid ideation brainstorming technique [9]. First, participants mapped all five jewellery roles on the whiteboard. Then, they individually ideated the concepts, values, techniques, features, and metaphors that would benefit individual roles. All ideas were collected (using post-it notes) and all group members familiarized with the input of others. Next, we moderated discussions to expand and explore all ideas. Participants were free to add new concepts to the board throughout the discussion or illustrate the ideas with quick sketches (see fig. 1). All workshop sessions were video- and audio-recorded, and the contents of whiteboards were photographed at the end of the session.

3.3 Analysis

We digitised photos of all participant notes, and transcribed audio footage verbatim. We appended participant-generated ideas with notes of relevant quotes (we used video footage to navigate relevancy of comments and their meaning within the discussion). A total of 196 participant-generated ideas formed our data corpus, with 104 notes generated during phase Q1, and 92 notes during phase Q2. Across Q1, 27 ideas were assigned to *identity* category, 23 to *prosperity*, 21 to *adornment*, 18 to *memento*, and least (15) to *protection* category. Consequently, for Q2 we collected 92 post-its: 33 in *identity*, 23 in *protection*, 17 in *adornment*, 14 in *memento* and 5 in *prosperity* category.

We decided to analyse our dataset using affinity diagramming [28], following the guidelines for contextual approach [6]. Since we grounded our inquiry in the existing framework, we analysed the data bottom-up. We looked for the emerging motives within each predefined category (adornment, identity, memento, prosperity and protection). We used physical post-its and a whiteboard during analysis. Then, we drew connections between categories and iterated our analysis for the whole dataset. Through the iterative process, post-its were regrouped and transferred until an overarching narrative describing the data has emerged. The diagramming process was conducted separately for subsets from Q1 and Q2 phases. Then, three researchers held iterative sessions to establish common themes for both phases.



Figure 1: During brainstorming, one participant sketched examples of ear cuff and septum ring to illustrate identity-expression with customisable, shape-changing and light-flashing pieces.

4 RESULTS

In this section, we reflect on the participants' answers to the two main questions that guided our inquiry: *Q1: What digital jewellery should be like, to support adornment, identity, memento, prosperity, and protection?*, and *Q2: Which digital functionalities can effectively support the socio-cultural roles of jewellery?* We examine participants' responses through the lens of Unger's framework [41]. Therefore, for each question, we present the results grouped according to the five roles defined by Unger. We illustrate our results with quotes from the recorded discussions, which are highlighted in italics and identified with participants' IDs.

4.1 Q1: What digital jewellery should be like, to support adornment, identity, memento, prosperity, and protection?

The first phase of the workshop brainstormed the ways in which interactive jewellery can effectively support various socio-cultural roles of wearable ornaments. This part focused on conceptual ideation, where participants contributed their views on meanings, metaphors, and emotions associated with interactive jewels. Participants were encouraged to contribute creative concepts, detaching from eventual implementation or economical constraints.

Adornment. Considering the aesthetics of modern jewellery, a vast majority of participants expressed their appreciation for designs that refer in their look to traditional jewellery pieces. A number of post-it notes directly indicated that such ornaments should maintain the outward appearance similar to the traditional pieces, both in form and materials used. One of the participants commented that modernistic appearance may trigger associations with intended uniqueness and explicit desire for self-expression.

Embracing the futuristic look would always lean more towards identity, since it [digital jewellery] is not yet widely adopted. (P10)

Considering potential interactive functionalities, our participants indicated that shape- and colour-changing objects would be particularly interesting. Participants imagined scenarios where digital jewellery dynamically adapted to the context of use, e.g. by matching its form to the outfit of the wearer. However, the experts highlighted the importance of embedding the interactive elements into the form factors of the pieces in a subtle, unobtrusive way. One of the participants emphasized that any technological intervention should bring direct functional benefits to be considered relevant and become appropriated as an everyday companion.

Adornment is efficiently expressed via traditional jewellery. We wear it because it is pretty. If we want to embed technology into it, it has to be utilitarian, carry an added value. (P13)

Identity. When discussing how digital jewels can support expression of the wearer's identity, our participants were especially enthusiastic about novel possibilities to personalise their pieces. Within the rich design space discussed, many ideas concerned various ways to express one's emotional or mental state via interactive features. The participants envisioned objects that would express the wearer's explicitly pre-defined state, or infer the output based on bio-signal tracking and ambient sensing. Further, the experts indicated that such jewellery would serve its role best when visible and conspicuous, encouraging the users to experiment with form factors and design metaphors. Digital jewellery for demarcating one's identity was often perceived on the symbolic level, as the participants found it enticing to explore new ways of encoding their preferences and emotional states in these wearable displays. One participant articulated the desire to challenge the limits of approval by experimenting with new designs.

There's more to it than allure. I enjoy cyberpunk-style of jewels. I want my appearance to challenge the fashion status quo, even if it would make someone uncomfortable. (P16)

Memento. Designing jewellery pieces that focus on supporting memento was perceived as promising and sparked lively discussions. Participants did not express the preferred appearance of this type of wearables, but cherished the possibility to upload personalised digital content onto them. One of the participants expressed the opinion that considering the role of a remembrance, its contents are more important than the form.

Memento is not about the item itself, but the meaning we associate to it. (P12)

Contrary to expressing identity, participants preferred the symbols used to be unclear to bystanders, restricting the possibility to decode the true meaning only to the wearer or intended persons. Participants suggested that it could be achieved by customising data representations in these artefacts.

The experts emphasized the new opportunities to embody significant memories in jewellery pieces in a personalised manner. However, they expressed concerns whether the wearable remembrances would be sufficiently long-lasting. While traditional memento-jewellery is often passed across generations, digital devices were often considered transient. Such perception of technology-augmented pieces was attributed to limited durability and proneness towards permanent damage caused by wear and environmental factors.

Prosperity. The experts expressed their reservations on whether digital jewellery can successfully act as a symbol of social and economic status. In their responses, they noticed that pieces made with precious metals and gems, often unique and of high monetary value, are likely to be more efficient in signifying one's prosperity. The discussions revealed that even expensive electronic gadgetry does not offer sufficient durability, and lacks uniqueness to be considered a truly exclusive object. Digital devices were often depreciated due to being mass-produced, in contrast to traditional jewellery

being perceived as an exhibition of craftsmanship of the jeweller. Further, unlike traditional pieces, electronic gadgets do not act as capital deposits and their value declines over time. While precious jewellery is often associated with being time-resistant, technology was perceived as quickly expendable.

On the other hand, the participants ideated around an alternative concept, where prosperity could be expressed by pioneering in innovations. Such jewellery would employ expensive, state-of-art technologies and be sold by premium and recognisable brands. Yet, the experts noted that the issue of a short life span remains unsolved, as software and hardware solutions outdate quickly.

Protection. Workshop participants often associated the role of protection with assumed religious or ethnic relevance. As a result, the experts were cautious in proposing novel ways of digitizing such pieces, emphasizing the potential risk of offending one's traditions or beliefs. The realm of beliefs and convictions was seen as inappropriate for experimentation with technology, which could potentially imply deprecating spiritually relevant artefacts. One participant expressed their concerns about inappropriate use of digital technology in wearable pieces.

I would be cautious when designing digital jewellery supposed to act as a protective amulet, as it is often associated with religious artefacts. For example, flashing LEDs in such pieces feels inappropriate to me. You could offend someone's religious feelings. (P4)

Interestingly, the experts were particularly interested in the literal meaning of protection. They observed that digital jewellery is capable to offer "protection" to the user by a number of means, e.g. monitoring their vital functions, triggering an alarm, or by recognising and mitigating stress. While the discussion did not focus on the preferred appearance of protective jewellery, one participant noted that unnoticeable integration of safety features might prove both beneficial and socially acceptable among different user groups.

Digital jewellery which would appear as its traditional substitute would encourage more users to wear it and potentially save their lives. For example, if you would like to give such a piece to your elderly family member, they could refuse to wear futuristic-looking pieces. (P2)

4.2 Q2: Which digital functionalities support adornment, identity, memento, prosperity and protection?

The second phase of the workshop focused on ideating particular interactive functionalities that would support different motivations to wear jewellery. Participants elaborated on specific interaction techniques, blending modalities and incorporating features known from other wearables and beyond that could benefit the socio-cultural dimensions of novel jewellery pieces.

Adornment. Participants identified various types of visual ornaments most suitable for enhancing the role of adornment. The proposed ideas ranged between incorporating simple light signals using LEDs, and compound visuals displayed on electrochromic or textile displays.

The experts suggested exploring various sensing techniques to make the objects context aware and allow for self-adapting behaviour. Discussed examples concerned altering the colour of a piece of jewellery in a way that it matches the rest of the outfit, or adapting to the weather conditions or current season. On the other hand, 2 participants opposed these ideas, emphasising their preference to control the appearance of their bijouterie manually. Moreover, a number of participants approved only these functionalities that could be hidden seamlessly inside the jewellery piece. Although the experts did not discuss any specific utilitarian benefits, they noticed the necessity to incorporate the electronics seamlessly, as visible wires or batteries would significantly impair the decorative potential of these ornaments.

Identity. The idea of incorporating different visual representations such as light manipulations with the use of RGB LEDs were frequently mentioned on the collected post-its and during the discussions. Other features that were deemed particularly promising considered various approaches to shape-changing or reconfigurable pieces. Among digital functionalities mentioned, the answers focused on allowing for richer personalisation and customisation. A few experts advocated for using various sensors for tracking bio-signals or incorporating context awareness. This way, the appearance of individual pieces would respond dynamically to the location, physical or mental state of its wearer. Introducing a degree of autonomy to the system was considered especially interesting, by offering cues for self-reflection or increasing awareness of one's body.

Memento. When debating on how the digital jewellery can effectively act as remembrances, the participants ideated around using the pieces as data storage containing personally relevant information about their relatives or important life events. Lively discussions emerged around how such remembrance should be expressed by a particular piece. Few participants suggested employing audio interfaces, offering the playback of voice recordings or tunes of personal relevance. Further, the experts digressed about saving location data related to important life events, or storing personal multimedia footage. Interestingly, participants struggled to ideate effective ways of direct access to such data, that would couple usability with sufficient protection of sensitive, private contents.

Prosperity. Ideating interactive modalities that would effectively act as status symbols proved especially challenging. The participants often referred to premium smartwatches as those which pursue similar prestige as their traditional counterparts. During the discussion, major concerns were raised on whether expressing status through technology is socially relevant. One participant directly expressed their scepticism towards expressing social status via interactive jewellery.

The value of electronics is in its innovation. As soon as some functions become popular and accessible, the item devaluates, while e.g. a gold ring will always remain precious. (P2)

Protection. Inline with observations from phase Q1, participants approached the protective role of jewellery literally. Digital functionalities associated with this role focused on self-monitoring and

raising alarms in case of emergency. Participants listed audio-based alarms (warning either the user or passers-by), location sharing, bio-tracking (e.g. heart rate, temperature, blood oxygen saturation, etc.) or ambient sensing (e.g. temperature, UV index, humidity, etc.) as potentially relevant features.

5 DISCUSSION

Here, we synthesize our insights and discuss implications of our work for future designs of digital jewellery. We reflect on our observations through the lens of traditional jewellery's roles described by Unger [41] (see section 2.2) and recent research endeavours on interactive jewellery.

5.1 Personalising Digital Jewellery

Throughout the workshops, the possibility to personalise and customise digital jewellery pieces was deemed most important. This manifested itself the most in the role of expressing one's identity. In terms of traditional jewellery, identity has usually been expressed with the use of recognisable symbols associated with particular social groups or beliefs [41, 131-142], while digital technology offers a broad area of personalisation features which allow the user to express their individual characteristics and identity [33]. Our inquiry revealed that participants were eager to express their emotional, mental and physical state via interactive jewellery, although they preferred to do it in ambiguous ways, coding these meaning in the performance of digital functionalities incorporated into bijouterie. Interestingly, participants perceived the uncontrollable nature of such sensor-based visualisations as desirable. We observed that participants were willing to compromise uniqueness of digital jewellery pieces (as such devices would probably be mass-produced which is not often the case with traditional jewellery) for rich personalisation-related features.

As current commercial wearables still support limited customisation [21], we advocate that future design inquiries, especially those focused on identity expressions in particular, should explore different forms of personalisation in digital jewellery.

5.2 Protective Role - From Metaphor to Function

Our study showed that the protective role of digital jewellery is the most distant from the traditional view on this role described in Unger's work [41, p.20-22], where perceived security ensured by wearing a piece of ornamentation was understood metaphorically. Participants associated protective amulets with spirituality, while considered digital jewellery being rather profane. Therefore, our study revealed the sense of inappropriateness expressed by participants. It resulted in the shift in perception on protective function of digital jewellery – we observed that rather than symbolically, users require of digital jewellery to guard the wearer literally. Experts ideated on employing various alarms, health-monitoring functions and features allowing to call medical professionals in case of emergency. A number of previous studies investigated devices that served similar roles [27, 36], yet, we believe that looking at user-protecting wearable devices from the jewellery-perspective can further expand the knowledge on how to design them effectively. To ensure that digital security-supporting jewellery respects

spiritual beliefs of the wearer, we campaign that future designs focus on utilitarian functionalities which allow the user to monitor their long-term well-being or intervene in dangerous situations. Further discussion on when digital jewellery is still deemed jewellery versus a well-designed functional wearable will be necessary.

5.3 Perceived Technical Limitations of Digital Jewellery.

During the workshops, participants expressed various concerns regarding technical aspects of digital jewellery. We observed that these concerns picture a mental model of digital jewellery artefacts as similar to lo-fi prototypes. Especially during the discussion about jewellery as the indicator of prosperity, we noted that participants question the durability and longevity. Although, these problems are essentially important, technological progress solved them to the extent, where a digital wearable piece is not more fragile than many traditional craftworks [7, 32]. Therefore, we conclude that participants see digital jewellery as interesting gadgets which require to compensate somewhat obscure appearance with new enjoyable possibilities.

To address these concerns, we recommend that designers should openly communicate that currently available technology in combination with skilful design and implementation allow for high-fidelity digital jewellery artefacts.

5.4 Future Work and Limitations

We focused on preliminary exploration of different socio-cultural roles that digital jewellery can serve. We consider it an early step to establishing a coherent design guidelines for digital substitutes of jewellery. We aim to empirically investigate our findings for each of the described roles with non-expert participants to increase generalisability of our findings. Similarly to previous endeavours [25], we aim to obtain artefacts through the participatory design process. Finally, we foresee that these artefacts could be applied 'in the wild' [24] and tested for social perception and wide public acceptance.

While we designed our exploratory studies with the utmost care, we acknowledge that certain design decisions in our work may have introduced biases. First, due to the specific research sample, which consists of participants from central Europe, our conclusions do not take into consideration members of other, non-western cultures. The fact that participants were also specialists, affects the potential generalisability of the insights. According to Nielsen [31], they can act as "double experts" with their expertise covering the kind of interface being evaluated, as well as its users, but still, particularly creative or extraordinary ideas might have been missed. Moreover, although the framework which inspired our inquiry has been chosen attentively, we recognise that it does not cover all aspects of jewellery.

6 CONCLUSION

In our work, we present design investigations on how novel interactive jewellery could effectively support socio-cultural roles served by the traditional ornament pieces. Drawing on participatory design techniques, we conducted a series of exploration workshops with N=16 experts with various design-related backgrounds. In

our inquiry, we analysed the design space of interactive jewellery through the lens of a design framework defining key roles and motivations to wear traditional jewellery [41]. Our results show that different socio-cultural roles require individual design approaches. Moreover, we found that expressing identity via digital jewellery is achieved most efficiently by allowing the user for personalisation of wearable pieces. Our work contributes actionable insights concerning how interactive features of jewellery pieces can efficiently address user requirements originating in the rich socio-cultural context of decorative wearables.

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